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Structured Cybersecurity Framework for Critical Infrastructure Systems: Implementation of Core Security Modules

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Abstract—Critical Infrastructure (CI) systems such as power grids, water treatment facilities, and transportation networks are increasingly exposed to sophisticated cyber threats as they integrate with the Internet under Industry 4.0 paradigms. Traditional perimeter-based defenses are insufficient against modern attack vectors targeting Industrial Control Systems (ICS) and SCADA environments. This paper presents the design and implementation of a structured cybersecurity framework addressing five core security pillars: Confidentiality, Integrity, Authentication, Authorization, and Non-Repudiation. Two core modules are implemented and validated: (1) a Secure Access Module employing bcrypt password hashing with Time-based One-Time Password (TOTP) Multi-Factor Authentication (MFA) and Role-Based Access Control (RBAC), and (2) a Cryptographic Engine implementing AES-256-GCM authenticated encryption combined with ECDSA digital signatures over the NIST P-256 curve. All 12 unit test cases pass, confirming correctness and security under both normal and adversarial conditions.

Index Terms—Critical Infrastructure, Cybersecurity, AES-GCM, ECDSA, RBAC, Multi-Factor Authentication, SCADA, ICS Security.

I. INTRODUCTION

Critical Infrastructure (CI) systems—including power grids, water treatment facilities, and transportation networks—form the backbone of national security and economic stability. As these systems transition toward Industry 4.0, integration of Industrial Control Systems (ICS) with public networks has exposed them to sophisticated cyber threats, including command injection, man-in-the-middle attacks, and unauthorized SCADA node access.

Traditional security models focus primarily on perimeter defense, which is insufficient against modern threats requiring layered, in-depth protection. A structured framework must ensure secure access, continuous monitoring, rapid incident response, and reliable system recovery. This paper presents such a framework, leveraging cryptographic primitives to protect critical infrastructure from unauthorized access, data manipulation, and service disruption.

The contributions of this paper are as follows:

- Design of a layered cybersecurity architecture addressing all five information security pillars.
- Implementation of a Secure Access Module with bcrypt-based MFA and RBAC.
- Implementation of a Cryptographic Engine using AES-256-GCM and ECDSA-P256.
- End-to-end integration demonstrating secure command dispatch to SCADA nodes.
- Validation through 12 unit test cases covering normal and adversarial scenarios.

The remainder of this paper is organized as follows. Section II defines core security objectives. Section III surveys related work. Section IV identifies research gaps. Section V presents the system design. Section VI details the implementation. Section VII covers integration testing. Section VIII presents test results. Section IX concludes.

II. CORE SECURITY OBJECTIVES

The proposed framework satisfies five fundamental pillars of information security:

- **Confidentiality**—AES-256 encryption protects sensitive operational data from unauthorized disclosure.
- **Integrity**—SHA-256 HMACs and AES-GCM authentication tags ensure control commands remain unaltered in transit.
- **Authentication**—MFA combining bcrypt-hashed passwords with TOTP tokens verifies operator identity.
- **Authorization**—RBAC enforces the Principle of Least Privilege based on operator roles.
- **Non-Repudiation**—ECDSA digital signatures and audit logs provide accountability for all control actions.

III. LITERATURE SURVEY

A comprehensive survey of IEEE, Springer, and Scopus-indexed research was conducted to evaluate existing defense mechanisms for critical infrastructure.

A. Related Work

Kumar et al. [1] proposed securing smart grids via hybrid cryptography combining ECC for key exchange and AES-256 for data encryption. While effective, the solution suffers from high latency in real-time operations and lacks an automated recovery module.

Chen [2] introduced a Zero Trust Architecture (ZTA) for ICS environments using MFA and Software Defined Perimeters (SDP). However, the work does not address data integrity for legacy field devices operating on older protocols.

TABLE 1
SUMMARY OF LITERATURE REVIEW

Paper	Techniques	Gaps Identified
Kumar et al. (2024)	ECC + AES-256	High latency; no recovery module
Chen (2025)	MFA; SDP	No legacy device integrity
Blockchain CI (2023)	ECDSA; DLT	High cost for PLCs

A blockchain-based non-repudiation approach for water system command injection [3] employed ECDSA and distributed ledgers but incurs high computational cost on low-power PLC devices, limiting practical deployability.

B. Summary

Table 1 summarizes the surveyed literature. While individual cryptographic tools are effective in isolation, no existing work integrates secure access control, authenticated encryption, and automated incident recovery into a single unified framework suitable for real-world ICS environments.

IV. RESEARCH GAP

Three key research gaps were identified based on the literature:

- **Integration Gap:** No unified solution combines secure access control with cryptographic data protection and automated recovery for ICS environments.
- **Legacy Support:** Minimal consideration is given to resource-constrained PLC/SCADA hardware, limiting applicability of computationally intensive schemes.
- **Passive Monitoring:** Current systems detect anomalies but lack automated mitigation and real-time recovery protocols.

This work addresses the integration gap by combining a Secure Access Module and a Cryptographic Engine into a cohesive SCADA command dispatch pipeline.

V. PROPOSED SYSTEM DESIGN

A. System Architecture

The proposed framework consists of six layered components as shown in Fig. 1. An operator first passes through an Authentication Gateway; authorized commands then traverse the RBAC Policy Engine, Cryptographic Engine, and SIEM monitor before reaching the Critical Infrastructure layer.

B. Architecture Components

Secure Access Module handles MFA and RBAC, ensuring only authorized operators and M2M nodes interact with the system.

Cryptographic Engine implements AES-GCM for authenticated encryption and ECDSA for digital signatures, enforcing confidentiality, integrity, and non-repudiation.

SIEM collects system and network logs; SHA-256 hashing is applied per entry to detect tampering in real time.

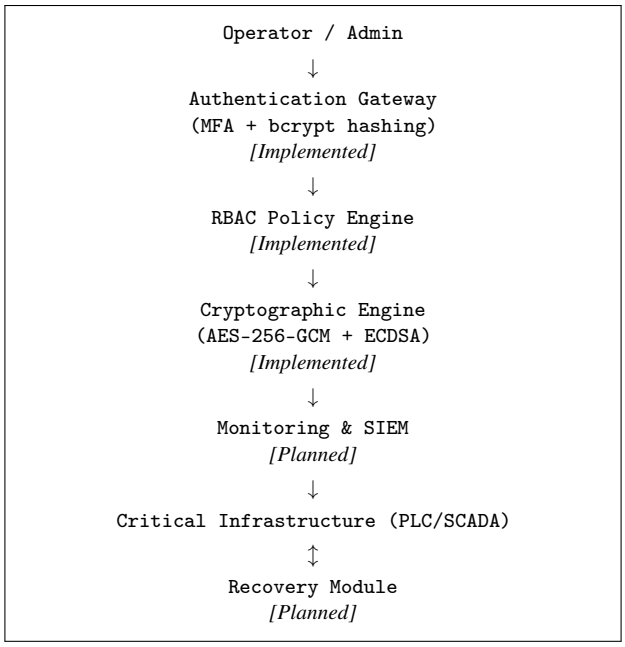

Figure 1. Architecture of the proposed cybersecurity framework. Green-shaded modules implemented in this work.

TABLE 2
DEVELOPMENT ENVIRONMENT

Component	Details
Language	Python 3.11
Cryptographic Library	cryptography v41.0 (PyCa)
Password Hashing	bcrypt v4.0
Operating System	Ubuntu 22.04 LTS / Windows 11
IDE	Visual Studio Code
Version Control	Git / GitHub
Testing Framework	unittest (Python standard library)

Recovery Module (planned) isolates compromised subnetworks and restores systems from verified golden-image backups upon breach detection.

C. System Workflow

1. Operator submits credentials; the Gateway verifies via bcrypt and TOTP.
2. RBAC Policy Engine grants or denies access by role.
3. Authorized commands are ECDSA-signed, then AES-GCM encrypted.
4. SIEM verifies SHA-256 digests on all packets.
5. On anomaly detection, the Recovery module isolates the affected sub-network.

VI. IMPLEMENTATION

Both modules are written in Python 3.11 using the `cryptography` library (v41.0) and `bcrypt` (v4.0). The development environment is summarized in Table 2.

TABLE 3
RBAC ROLE DEFINITIONS

Role	Permitted Operations
ADMIN	READ, WRITE, EXECUTE, SHUTDOWN, CONFIG
OPERATOR	READ, WRITE, EXECUTE
VIEWER	READ only

TABLE 4
CRYPTOGRAPHIC PARAMETERS

Parameter	Value
Encryption Algorithm	AES-256-GCM
Key Length	256 bits (32 bytes)
Nonce/IV Length	96 bits (12 bytes, random/msg)
Authentication Tag	128 bits (16 bytes)
Digital Signature	ECDSA, NIST P-256
Signature Hash	SHA-256
Key Derivation	HKDF-SHA256

A. Module 1 – Secure Access Module

a) Overview: The module provides two functions: (1) identity verification via bcrypt + TOTP MFA, and (2) RBAC enforcement restricting operations by assigned role.

b) RBAC Role Definitions: Three roles are defined following the Principle of Least Privilege (Table 3).

c) Authentication Mechanism: Password hashing uses bcrypt with work factor 12, providing resistance to offline brute-force attacks. TOTP generation employs HMAC-SHA256 over 30-second time windows per RFC 6238 [8]. Constant-time comparison via `hmac.compare_digest` prevents timing-oracle attacks.

```python
def authenticate(username, password, totp):
    user = USER_STORE.get(username)
    if not user:
        return None
    # Factor 1: bcrypt constant-time check
    if not bcrypt.checkpw(
            password.encode(),
            user.password_hash):
        return None
    # Factor 2: TOTP verification
    expected = _generate_totp(user.totp_secret)
    if not hmac.compare_digest(expected, totp):
        return None
    return user

def authorize(user, operation):
    permitted = ROLE_PERMISSIONS.get(
        user.role, set())
    return operation in permitted
```

Listing 1. Authentication and RBAC core logic (auth_module.py).

B. Module 2 – Cryptographic Engine

a) Cryptographic Parameters: Table 4 lists the cryptographic parameters used by the engine.

b) AES-256-GCM Encryption: AES-GCM simultaneously provides confidentiality and integrity via a 128-bit authentication tag. A fresh 96-bit nonce is generated per message using a cryptographically secure PRNG. Additional Authenticated Data (AAD) binds the device identifier to the ciphertext without encrypting it, preventing cross-node replay attacks.

c) ECDSA Digital Signatures: ECDSA over NIST P-256 provides non-repudiation for every control command. Each command is signed before encryption; the receiver verifies the signature after decryption. A tamper-detection test confirms that modifying even a single byte of the command causes signature verification to fail.

```python
def encrypt_aes_gcm(key, plaintext, aad=b""):
    nonce = os.urandom(12)   # 96-bit random nonce
    aesgcm = AESGCM(key)
    ciphertext = aesgcm.encrypt(
        nonce, plaintext, aad or None)
    return nonce, ciphertext

def sign_command(private_key, command):
    return private_key.sign(
        command, ec.ECDSA(hashes.SHA256()))

def verify_signature(public_key, command, sig):
    try:
        public_key.verify(
            sig, command,
            ec.ECDSA(hashes.SHA256()))
        return True
    except InvalidSignature:
        return False
```

Listing 2. Cryptographic Engine core functions (crypto_engine.py).

VII. INTEGRATION: SECURE COMMAND DISPATCH

The two modules are integrated into a secure command dispatch pipeline for SCADA environments. The end-to-end workflow proceeds through six steps:

1. **Authentication:** Operator credentials and TOTP verified via bcrypt and HMAC-SHA256.
2. **Authorization:** Role evaluated; WRITE access granted for OPERATOR/ADMIN.
3. **Sign:** Control command signed using ECDSA-P256.
4. **Encrypt:** Command encrypted with AES-256-GCM; device ID bound as AAD.
5. **Dispatch:** Packet `[nonce || ciphertext+tag || ECDSA_sig]` transmitted to the SCADA node.
6. **Verify:** Node verifies ECDSA signature and AES-GCM authentication tag before executing.

The integration test confirms all five security goals are satisfied simultaneously:

```
[+] Secure command dispatch SUCCESSFUL.
Goals: Confidentiality | Integrity |
Authentication | Authorization |
Non-Repudiation
```

VIII. TESTING AND VALIDATION

Unit tests were written using Python's built-in `unittest` framework. Table 5 summarizes all 12 test cases and their outcomes.

TABLE 5
TEST CASES AND RESULTS

TC#	Description	Exp.	Act.
TC01	Valid credentials + TOTP	PASS	PASS
TC02	Incorrect password	FAIL	FAIL
TC03	Wrong TOTP token	FAIL	FAIL
TC04	Unknown username	FAIL	FAIL
TC05	ADMIN requests SHUTDOWN	GRANT	GRANT
TC06	VIEWER requests WRITE	DENY	DENY
TC07	OPERATOR requests CONFIG	DENY	DENY
TC08	AES-GCM encrypt/decrypt	OK	OK
TC09	Tampered AES-GCM ciphertext	INVALID	INVALID
TC10	ECDSA sign + verify	VALID	VALID
TC11	ECDSA tampered command	INVALID	INVALID
TC12	ECDSA wrong public key	INVALID	INVALID
All 12 / 12 test cases passed			

TABLE 6
SECURITY GOALS AND IMPLEMENTATION STATUS

Goal	Approach	Status
Confidentiality	AES-256-GCM	Implemented
Integrity	AES-GCM 128-bit tag	Implemented
Authentication	bcrypt + TOTP MFA	Implemented
Authorization	RBAC policy engine	Implemented
Non-Repudiation	ECDSA-SHA256	Implemented
Monitoring (SIEM)	SHA-256 log hashing	In Progress
Recovery	Automated isolation+restore	In Progress

All adversarial cases (TC02, TC03, TC06, TC07, TC09, TC11, TC12) correctly reject invalid or tampered inputs, confirming robustness against brute-force, replay, privilege escalation, and tampering attacks. Table 6 maps each security goal to its implementation status.

AES-256-GCM provides 256-bit symmetric security, while ECDSA over P-256 provides approximately 128-bit security—both exceeding current NIST recommendations [4,5]. The bcrypt work factor of 12 requires approximately 250 ms per hash attempt, making online brute-force attacks computationally infeasible.

IX. CONCLUSION

This paper presented the design and implementation of a structured cybersecurity framework for Critical Infrastructure Systems. Two core modules were implemented and validated: the Secure Access Module combining bcrypt-based MFA with RBAC, and the Cryptographic Engine employing AES-256-GCM and ECDSA-P256 for authenticated encryption and digital signing of all SCADA command traffic.

All 12 unit test cases pass under both normal and adversarial conditions, confirming that the framework simultaneously satisfies Confidentiality, Integrity, Authentication, Authorization, and Non-Repudiation. The end-to-end integration demonstrates a complete secure command dispatch pipeline from operator authentication to SCADA node execution.

Future work will address Continuous Monitoring via SHA-256 log integrity chains (SIEM) and Automated Incident Response and Recovery using golden-image restoration. Performance benchmarking on resource-constrained PLC hardware will also be conducted to validate practical deployability.

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